

OSTEOMYELITIS

Inflammation of bone and marrow, secondary to bacterial, viral, fungal or parasitic infection.

No specific organism is identified in nearly 50% of patients.

Pyogenic osteomyelitis - bacterial infection:

- hematogenous spread
- extension from a contiguous site
- direct implantation (e.g., in traumatic injury)

Staphylococcus aureus: 80% to 90% of culture-positive pyogenic osteomyelitis

Escherichia coli , **Pseudomonas** , and **Klebsiella**

Haemophilus influenzae and **group B streptococci** : neonates

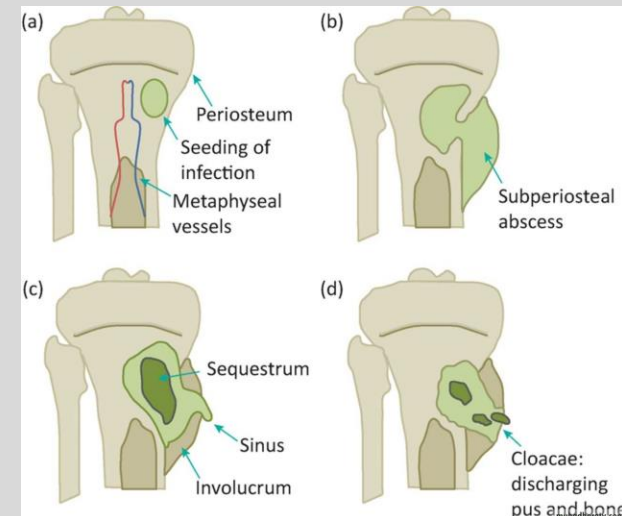
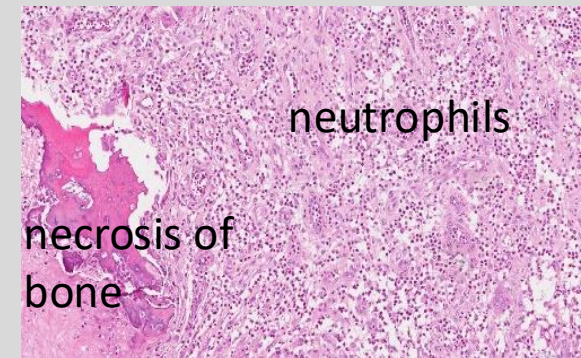
Sickle cell disease predisposes to Salmonella infection

OSTEOMYELITIS

- **ACUTE:** bacteria proliferate and recruit neutrophils
within 48 hours: necrosis of bone cells and marrow
bacteria: periosteum → **subperiosteal abscess**
- → **periosteal lifting further impairs blood supply**
- → **necrosis**
- → **soft tissue abscesses after periosteal rupture – open to skin as draining sinuses**
- Dead bone, or **sequestrum** is formed and this structure can crumble and release fragments into sinus tract forming a so called CLOACAE indicating a **CHRONIC** Stage; this is seen in 5% to 25% of cases
- As inflammation continues, chronic inflammatory cells recruited during first week release cytokines that stimulate bone resorption, fibrous tissue ingrowth, and peripheral deposition of reactive bone; **this new bone can form a living shell, or involucrum, around devitalized, infected bone**
- **Xray:** lytic focus of bone destruction surrounded by sclerosis

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SERONEGATIVE SPONDYLOARTHROPATHIES

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Undefined microbial antigens  cross-react with musculoskeletal
system components  T-cell responses 
Immune Mediated Seronegative Spondyloarthropaty

Heterogeneous group of disorders

- Absence of rheumatoid factor
- Pathologic changes in ligamentous attachments rather than synovium
- Sacroiliac joint involvement
- Association with **HLA-B27**
- Bony proliferation leading to ankylosis

Examples: Ankylosing spondylitis, infectious arthritis, reactive arthritis

REACTIVE ARTHRITIS (REITER'S SYNDROME)

TRIAD: arthritis, nongonococcal urethritis or cervicitis, and conjunctivitis

- Mono- or oligoarticular arthritis
- Intestinal infections: Salmonella, Shigella, Campylobacter
- STD: Chlamydia trachomatis
- Rarely: group A streptococcal infections
- RF negative; HLA-B27 positive
- young adults
- 50%: Recurrent Arthritis, Tendonitis, Lumbosacral pain;
- Active disease > 6 months: **CHRONIC REACTIVE ARTHRITIS**



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Presentation:

Initial acute-onset asymmetric oligo-arthritis (knee) often with inflammatory low back pain. In some patients: Inflammation at insertions of Achilles tendon and plantar fascia with heel swelling; inflammation of digits (dactylitis), joint stiffness, and low back pain.

Extraarticular :

- conjunctivitis and uveitis;
- dysuria, pelvic pain, urethritis,
- balanitis, cervicitis, mucosal ulcers,
- skin rashes, and cardiac valvular disease

LYME ARTHRITIS

Agent: Spirochete *Borrelia burgdorferi*

Deer ticks (*Ixodes*);

Leading arthropod-borne disease in US

New England, mid-Atlantic states, and upper Midwest

Classic Lyme disease – involves multiple organ systems; and has three clinical phases:

Early localized stage: initial skin infection, followed (days/weeks) by dissemination of organism to other cutaneous sites, cranial nerves, heart, and meninges

➡ **Early disseminated stage**

The untreated **or the late disseminated disease** is associated with a migratory arthritis (**Lyme arthritis**) in 80 % of cases (**months after infection, lasts weeks to months**)

Arthritis: 10% of *Borrelia* infections

Most are treated and cured at earlier stage

Joints: knees, shoulders, elbows, and ankles
(one or two joints)

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- **Rash- Lyme rash:** bull's-eye with circles around the middle. However, most are round, red, and at least 2 inches across.
- Grows to 10- 12 inches across within days.
- Warm to the touch
- Not itchy or pain
- Location: anywhere



FIBROUS DYSPLASIA

Benign tumor: all components of normal bone present → do not differentiate into mature structures;

During skeletal development .

Pathogenesis: various somatic gain-of-function mutations in *GNAS1* (encodes the stimulatory α subunit of G s, resulting in constitutively active G s protein) → promoting cellular proliferation and inhibition of osteoblast differentiation.

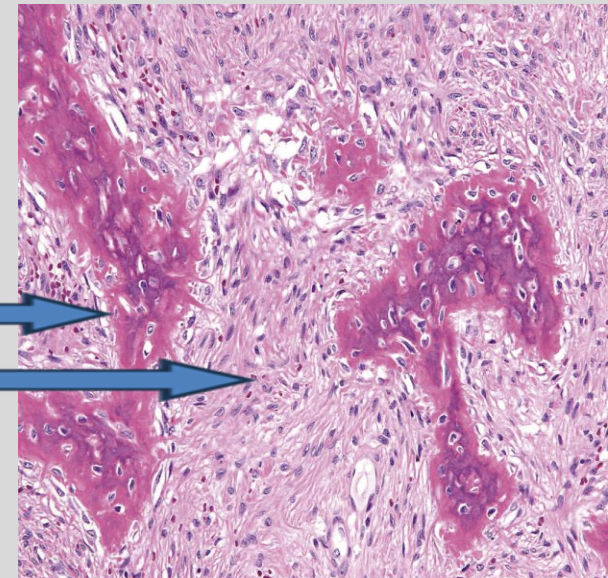
Intramedullary lytic lesions: bowing and cortical Bone thinning;

Histo:

curvilinear trabeculae of woven bone →
moderately cellular fibroblastic proliferation →
without prominent osteoblastic rimming

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FIBROUS DYSPLASIA

Clinical: pain, swelling, pathologic fractures, severe bone deformity

Monostotic fibrous dysplasia - early adolescence: femur, tibia, ribs, jawbones, and calvarium usually; asymptomatic, can cause pain, discrepancies in limb length, and pathologic fracture; treated by curettage, recurrence common

Polyostotic fibrous dysplasia - earlier age, developing into adulthood; femur, skull, and tibia; craniofacial involvement; progressive with crippling deformities, and fractures; bisphosphonates to reduce severity of bone pain; **rare malignant transformation into sarcoma**

McCune-Albright syndrome — precocious sexually development, esp. in girls; hyperthyroidism, pituitary adenomas that secrete growth hormone, and primary adrenal hyperplasia; unilateral; skin lesions (café-au-lait spot) on same side of body



Aneurysmal Bone Cyst (ABC)

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Multiloculated blood-filled spaces.; all ages affected, most in adolescence; often in femur, tibia, vertebral body

Xray: expansive, well-circumscribed lytic lesion with well-defined margins; most show central lysis and thin sclerotic “eggshell” of reactive bone at periphery
Multiple blood-filled cystic spaces separated by thin, tan-white septa composed of plump spindle cells, multinucleated osteoclast-like giant cells, and reactive woven bone lined by osteoblasts

70%: chromosome 17p13 rearrangements present within plump spindle cells → results in fusion of **USP6* coding region** with promoters of genes highly expressed in **osteoblasts** (CDH11)

Clinical: localized pain and swelling → limp, vertebral lesions, and nerve compression

Benign, but locally aggressive
treatment by curettage or excision with recurrence in 10% to 50% of cases



PRIMARY BONE TUMOURS

- Rare; outnumbered by metastases;
- Long bones of extremities
- Asymptomatic, identified incidentally
- Some are painful
- Slow-growing mass
- Pathologic fracture

- Important factors
 - Clinical Presentation
 - Radiographic Appearance
 - Age of Patient
 - Location of Lesion
 - Microscopic Appearance

During the first 3 decades of life—the ages of the greatest skeletal growth activity

Location: Distal femur and proximal tibia (bones with highest growth rate)

Relatively specific radiographic presentations

Benign >> Malignant

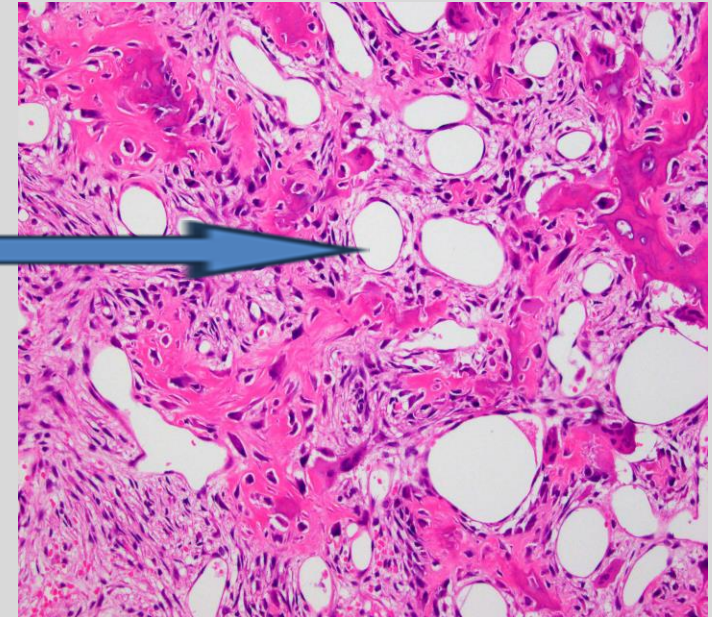
Age probably most important clinical clue to diagnosis

Painful lesions: growing lesions (malignancy or locally aggressive)

BENIGN TUMORS: OSTEOID OSTEOMA AND OSTEOLASTOMA

- Benign bone-producing tumors: **similar histologic features** but different sizes, sites of origin, and symptoms; malignant transformation rare
- Round-to-oval, well-circumscribed masses of hemorrhagic, gritty tan tissue
- **Micro:** trabeculae of woven bone rimmed by single layer of prominent osteoblasts, loose connective tissue with dilated and congested capillaries
- Reactive bone encircles actual neoplasm, or nidus (small mineralized radiolucent core)
- **Osteoid osteoma:** tx by radiofrequency ablation
- **Osteoblastoma:** curettage or en bloc excision

woven bone



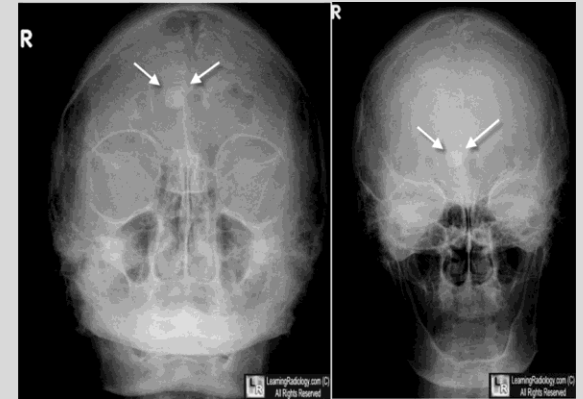
OSTEOID OSTEOMA

- Benign tumor of bone with no malignant potential
- Less than **< 2 cm in size**
- Males <25 years old; **long bones** and on the surface of facial bones or skull

Gardner's Syndrome:

Familial Adenomatous Polyposis + osteomas and soft tissue tumors

- Slow growing, no clinical significance unless : obstruction or cosmetic issues
- **Nocturnal bone pain** (secondary to excess prostaglandin E2 produced by proliferating osteoblast), that is **relieved by Aspirin and NSAIDS**
- **Clinical:** a thick rim of reactive cortical bone may be only investigational finding



CHONDROMA (ENCHONDROMA)

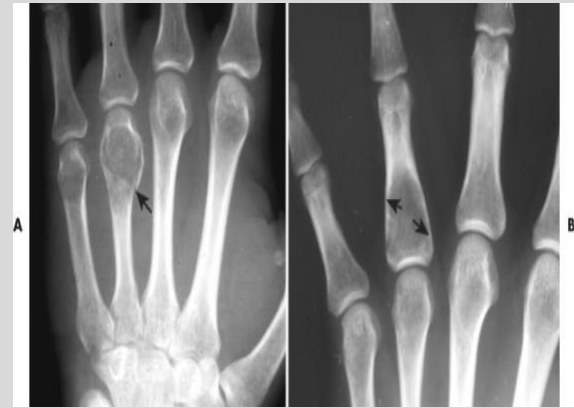
- **Benign tumor of hyaline cartilage** : within medullary cavity (enchondromas) or the on the bone surface (juxtacortical chondromas)
- The patients are 20 – 50 years old;
- Asymptomatic
- Painful; Pathological Fractures
- Well-circumscribed nodules usually < 3 cm
- Cut surface: gray-blue, and translucent
- Heterozygous mutations in IDH1 and IDH2 genes

Histology: hypocellular lesion with an abundance of hyaline cartilage. The chondrocytes appear normal. A peripheral endochondral ossification may be noted.

Xray: circumscribed lucency with central irregular calcifications, sclerotic rim, and intact cortex

Treatment: usually observation or curettage;

Solitary chondromas rarely undergo sarcomatous transformation, but those seen in Ollier disease and Maffucci syndrome do so more frequently



Ollier disease and Maffucci syndrome: nonhereditary disorders characterized by multiple enchondromas

Ollier disease: enchondromas confined to one side and limited to the limbs. Increased risk of chondrosarcoma 25-30% at 40 years . Increased risk of developing gliomas, pancreatic and juvenile granulosa cell tumors of the ovary.

Maffucci syndrome: multiple enchondromas and spindle cell hemangiomas. The patients are at risk of developing other malignancies, including brain gliomas

GIANT CELL TUMOR (Osteoclastoma)

- Rare benign tumor of multinucleated osteoclast-type giant cells; locally aggressive
- 4% metastasize to lungs or they may regress spontaneously
- rarely fatal
- knee, involving distal femur or proximal tibia
- Arthritis-like symptoms
- Pathologic fractures;
- May destroy overlying cortex, resulting in bulging soft tissue mass

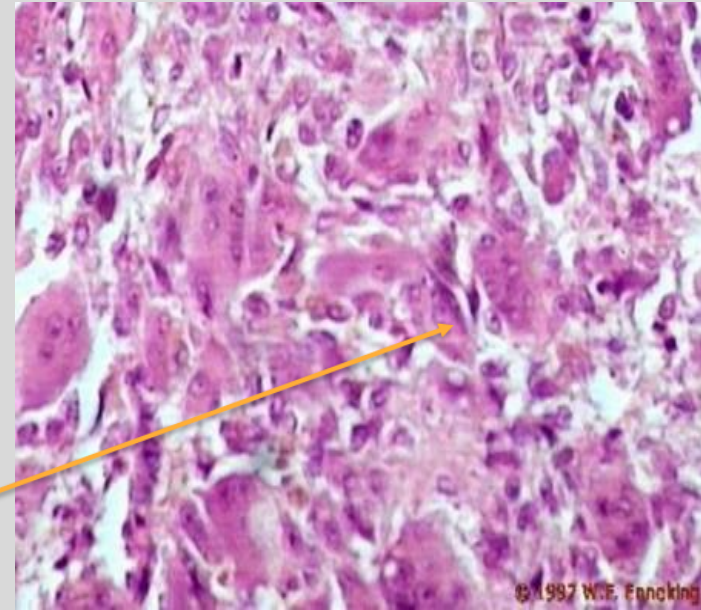
Macroscopically, these are large, red-brown masses that frequently undergo cystic degeneration

Histology shows uniform oval mononuclear tumor cells and abundant **osteoclast-type giant cells with 100 or more nuclei**;

- **Necrosis and mitotic activity may be prominent**

Xray: Soap Bubble

Therapy: curettage, 40% to 60% recur locally;



OSTEOSARCOMA MORPHOPATHOLOGY

Bulky, gritty, gray-white tumors, usually with hemorrhage and cystic degeneration; destroys surrounding cortices to produce soft tissue masses, spread extensively into medullary canal, replacing hematopoietic marrow

Histology: pleomorphism, large hyperchromatic nuclei, bizarre tumor giant cells, and abundant mitoses including abnormal (e.g., tripolar) forms; extensive necrosis and intravascular invasion common

Diagnosis: presence of malignant tumor cells producing unmineralized osteoid or mineralized bone

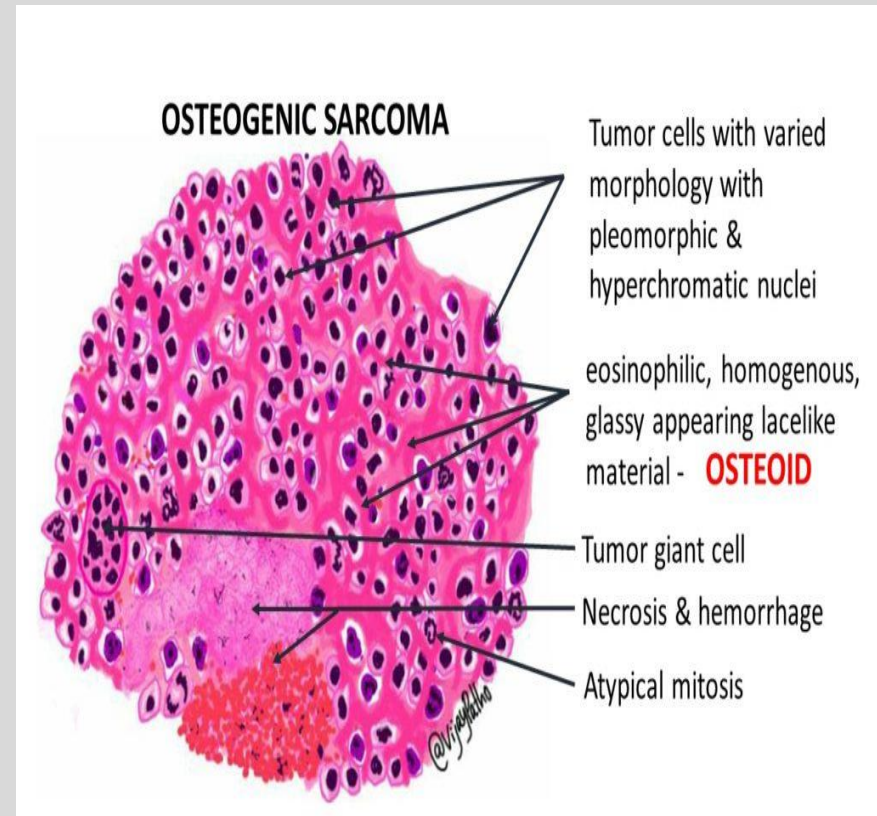
All cases: assumed to have occult metastases at time of diagnosis →

Hematogenous metastases :lungs, bones, brain, and other sites

5-year survival rate is less than 20% for those with metastases, recurrent disease, or secondary osteosarcoma

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OSTEOSARCOMA CLINICAL and GENETIC FEATURES

- Pain and Pathologic fractures
- Peak incidence: adolescent growth spurt;
- Location: in the growth plate of rapidly growing bones (as increased proliferation may predispose to mutations)
- Acquired genetic abnormalities including chromosomal aberrations; usually associated with mutations such as:
 - RB mutations in up to 70% of sporadic osteosarcomas;
 - Germline mutation of TP53 in individuals with Li-Fraumeni syndrome: sarcomas and (osteo)sarcoma, breast, brain, and adrenal cancer
 - CDKN2A (encodes two tumor suppressors, p16 and p14) inactivated in many osteosarcomas

Ewing Sarcoma

- Malignant bone tumor: undifferentiated primitive round cells without differentiation;
- 6% to 10% of primary malignant bone tumors;
- 2nd most common group of pediatric bone sarcomas
- 80% of patients <20 years of age;
- Boys > Girls with predilection for Caucasians;
Rarely African or Asian descent

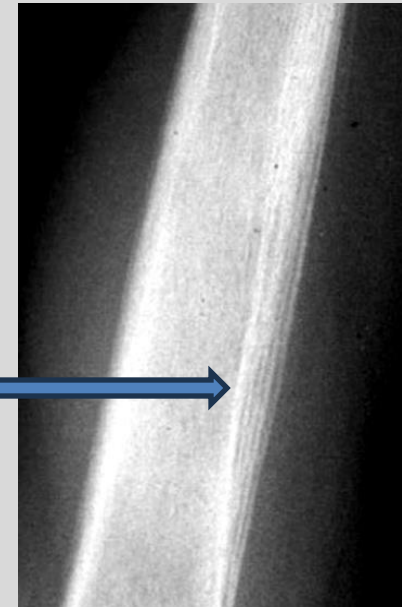
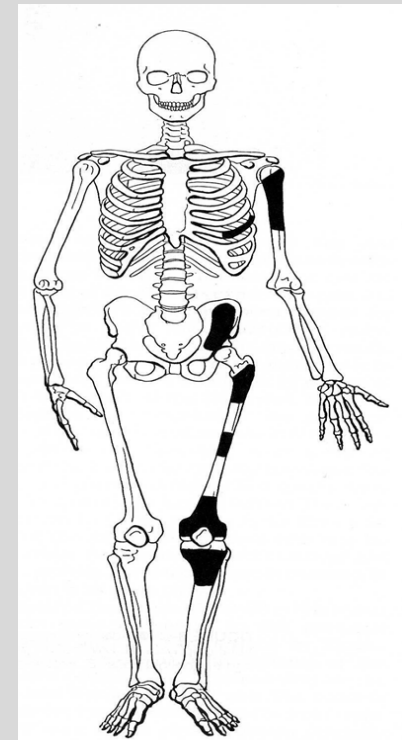
Location: Diaphysis of long tubular bones, femur and pelvis, humerus and ribs

Presentation:

- painful, enlarging mass
- affected site is frequently tender, warm, and swollen;
- systemic findings that mimic infection: fever, elevated ESR, anemia, and leukocytosis

Xray:

- lytic tumor with permeative/moth eaten margins
- periosteal reaction produces
- reactive bone deposited in an onion-skin fashion



t(11;22); > 90%: balanced translocation involving EWSR1 gene on chromosome 22 and FLI1 gene on chromosome 11



EWSR1/FLI1 fusion gene
encodes a chimeric EWS/FLI1 protein



binds to chromatin and dysregulates transcription



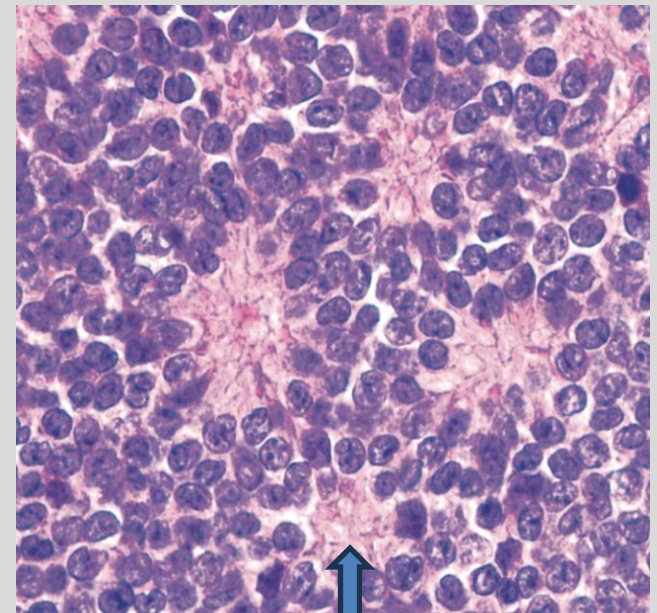
uncontrolled growth and abnormal differentiation

Origin: mesenchymal stem cells?
primitive neuroectodermal cells?

Starts in medullary cavity and invades cortex, periosteum, and soft tissue

Macro: Soft, tan-white, with areas of hemorrhage and necrosis

- Aggressive
- Neoadjuvant chemotherapy followed by surgical excision and adjuvant chemotherapy, with or without irradiation, has achieved 75% 5-year survival and 50% long-term cure rates



Micro: sheet of uniform small, round cells with scant cytoplasm

Homer-Wright rosettes: rounded cell clusters with a central fibrillary core, indicating a neuroectodermal differentiation.

CONVENTIONAL CHONDROSARCOMA

Bulky tumors:

- nodules of glistening gray-white, translucent cartilage,
- matrix is often gelatinous or myxoid,
- ooze from cut surface;
- spotty calcifications typically present;
- central necrosis may create cystic spaces

Histo:

- neoplastic cartilage infiltrates marrow space
- variable cellularity, pleomorphism,
- degree of cytologic atypia
- mitotic activity;

Grading from 1 to 3; **Histologic grade correlates directly with biologic behavior**

Most conventional chondrosarcomas are grade 1 tumors

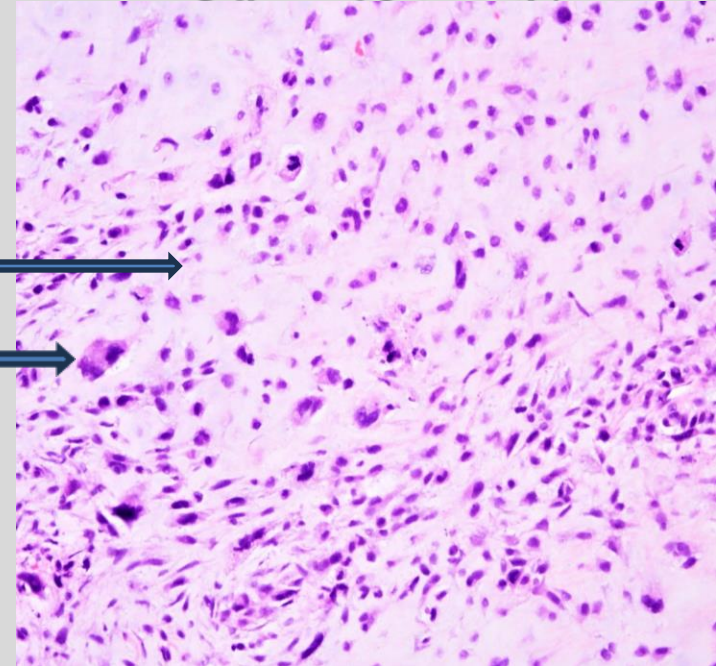
Rarely metastasize

5-year survival rates of 80% to 90%

70% of grade 3 tumors spread hematogenously, especially to lungs and the 5-year survival is 43%

Treatment: Conventional wide surgical excision; mesenchymal and dedifferentiated tumors (more aggressive): excision and adjuvant chemotherapy

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METASTATIC BONE TUMORS

Bidirectional interactions between metastatic cancer cells and native bone cells cause changes in bone matrix

Tumor cells do not directly resorb bone in lytic lesions, but secrete, prostaglandins, cytokines, and PTHrP



up-regulate RANKL on osteoblasts and stromal cells



stimulating osteoclast activity

Tumor cell growth supported by release of matrix-bound TGF- β , IGF-1, and FGF as bone is resorbed

Presence of bone metastases: poor prognosis

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Treatment - symptomatic relief of pain by analgesics and radiotherapy.

Prevention of further development of bone metastases by bisphosphonates
zoledronic acid